

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, offering over 4 years of experience as a Software Developer and Data Engineer. Possesses a Master of Science in Computer Science with expertise in building scalable microservices, distributed data pipelines, and implementing robust software design principles. Demonstrates strong command of Java, Python, Spark, SQL, and cloud platforms (AWS, Azure), coupled with experience in CI/CD automation, C++ development, and graduate-level work on large-scale data platforms and privacy-preserving AI. Proven ability to leverage version control and workflows for efficient software development.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, handling over 50,000 daily API requests with 99.99 percent SLA uptime, demonstrating strong software design.
- Engineered and automated CI/CD pipelines using Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments, improving workflows.
- Modernized legacy systems by migrating to Log4j 2.x and enhancing monitoring and error handling, reducing issue resolution time by 20 percent.
- Led a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing, practicing effective version control.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, focusing on robust software design.
- Improved system performance by 90 percent through SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, enhancing development workflows.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices, promoting C++ best practices where applicable.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage, a key aspect of software development.
- Supported release planning and sprint execution using JIRA within Agile development teams, leveraging effective workflows.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze more than 8 GB of NYC taxi trip data in near real time.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization, leveraging Python for software development.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments, integrating Go for certain workflow automation.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight.

Privacy-Preserving AI Email Client (Capstone Project - T5, LoRA, Python) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, demonstrating advanced software design.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, utilizing Python extensively.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, ensuring data structure integrity.

SKILLS

- Programming Languages: Proficient in Java and Python, with experience in C++ and Go for diverse software development needs.
- Frameworks and Libraries: Expertise in Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, and JUnit

for robust application development.

- Web Services: Skilled in designing and implementing REST API and SOAP API solutions, emphasizing efficient software design.
- Cloud Platforms: Experienced with AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), and Google Cloud Platform for scalable infrastructure.
- Databases: Proficient in PostgreSQL, Oracle Database, IBM DB2, and RDS, with strong SQL data manipulation skills.
- Tools and Technologies: Adept in Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for comprehensive software development and workflows.
- Concepts: Strong understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, and Privacy-preserving LLM-based applications.
- Data Analysis and Visualization: Skilled in Tableau and QuickSight for generating actionable insights, supported by strong data structure understanding.
- Core Competencies: Incorporating Version Control, Software Design, and effective Workflows into all aspects of software development.